

NABIL IBTEHAZ

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Links : [Homepage](#) ◇ [Google Scholar](#) ◇ [LinkedIn](#) ◇ [Github](#)

BACKGROUND

4th year PhD Student of Computer Science, with **7+ years** of experience in Deep Learning, specialized in Biological and Biomedical applications. My papers have received **2800+** citations and I proposed **7 US patents**.

EXPERIENCE

Futurewei Technologies, Santa Clara, CA, USA

Aug 2024 - May 2024 || Aug 2023 - May 2023

ML Algorithms Intern

Over the 2 summers, studied several problems on computer vision and multimodal learning, involving modal alignment and robustness against missing modalities. Published papers on **CVPR and ICLR workshops**.

Samsung R&D Institute, Dhaka, Bangladesh

Feb 2019 - Jul 2021

Software Engineer

Involved in a number of R&D projects related to smartphone camera application and health sensor data. Contributed to **4 A1 and 1 A2 patents**. Developed PoC for various AI applications for Samsung smartphone ecosystem.

Ensembl, EMBL-EBI, Cambridge, UK

May 2018 - Aug 2018

Summer Intern (under Google Summer of Code program)

Worked on a research project of refining orthologous relationships between genes to transcript level, [\[Project Link\]](#).

RECENT PUBLICATIONS

- **Ibtehz, Nabil**, et al. "Fusion of regional and sparse attention in Vision Transformers." *CVPR 2024 Workshop on Transformers for Vision (T4V)*
- **Ibtehz, Nabil**, Mortazavi, M. "Modally Reduced Representation Learning of Multi-Lead ECG Signals through Simultaneous Alignment and Reconstruction." *ICLR 2024 Workshop on Learning from Time Series For Health*.
- **Ibtehz, Nabil**, Kagaya, Yuki, and Kihara, Daisuke . "Domain-PFP allows protein function prediction using function-aware domain embedding representations." *Nature Communications Biology* 6.1 (2023): 1103.
- **Ibtehz, Nabil**, and Kihara, Daisuke . "Acc-unet: A completely convolutional unet model for the 2020s." *International Conference on Medical Image Computing and Computer-Assisted Intervention. MICCAI, 2023*.

SELECTED PATENTS

- Method and apparatus for monitoring physical activity (US20210279554A1, WO2021177694A1, KR20210111948A)
- Wearable device and method for controlling thereof (US20230229260A1, WO2022065682A1, KR20220040277A)
- Electronic apparatus and controlling method thereof (US20230205923A1, WO2022050558A1, KR20220029212A)

EDUCATION

PhD in Computer Science, Purdue University, USA (CGPA 3.83/4.00)

Aug 2021 -

Conducting PhD thesis on Biologically meaningful representation learning and LLM at [Kiharalab](#).

MSc in Computer Science and Engineering, BUET, Bangladesh (CGPA : 4.00/4.00)

Oct 2017 - Feb 2020

Thesis : Inferring health conditions through applying a fusion of machine learning and biomedical signal processing

SKILLS

Python, Jupyter, JavaScript, C++, Java, PyTorch, Tensorflow, Scikit-Learn, Angular, AWS, PostgreSQL
Computer Vision, LLM, Multimodal learning, Image processing, Signal processing
Protein bioinformatics, Medical imaging, Biomedical signal processing, Phylogenetics